

FarmerChat -- QA Automation Executive Summary

Project: FarmerChat Android (v4.0.0)

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What Was Done

1. Automation Suite Development

Built a comprehensive **end-to-end UI automation test suite** for the FarmerChat Android app using the **Maestro** framework.

Metric	Value
Total Test Cases	40
Reusable Helper Flows	9
Modules Covered	8
Priority P0 (Critical)	14 tests
Priority P1 (High)	16 tests
Priority P2 (Medium)	10 tests

Modules Covered:

Onboarding | Home Feed | AI Chat | Settings | Authentication | Help & Support | Navigation | Error Handling

2. CI/CD Pipeline Setup

Platform	Purpose	Status
Jenkins (Local)	Automated pipeline with HTML reports + video recordings	Operational
BrowserStack (Cloud)	Real device testing via REST API integration	Operational

3. Test Execution on Devices

#	Device	Type	Platform	Where
1	Pixel 7 Pro (API 34)	Emulator	Android 14	Local / Jenkins
2	Samsung Galaxy S24	Real Device	Android 14	BrowserStack Cloud

Total devices tested: 2 (1 emulator + 1 real device)

BrowserStack Real Device Results

Device: Samsung Galaxy S24 (Android 14)

Tests Run: 35 out of 40

Result	Count	Percentage
Passed	32	91.4%
Failed	3	8.6%
Skipped	0	0%

3 Failed Tests -- Root Cause

Test	Failure Reason	App Bug?
Home Screen Verification	Server did not return "Learn More" feed card	No -- server content
Share AI Response	Share sheet text differs on Samsung vs emulator	No -- device UI difference
Save AI Response	Confirmation toast disappeared too quickly	No -- timing sensitivity

All 3 failures are environment-related, not application defects.

5 Tests Excluded from BrowserStack

Test	Reason
Photo Query (Camera)	Requires camera hardware simulation
Error Screen Retry	Requires airplane mode toggle (ADB only)
Chat Error Recovery	Requires network manipulation (ADB only)
Chat History Retry	Requires network manipulation (ADB only)

Test	Reason
Photo Query (Gallery)	Requires pre-loaded gallery image (ADB only)

These 5 tests run successfully on the local emulator via Jenkins.

Key Deliverables

#	Deliverable	Description
1	40 Maestro test flows	Covering all critical user journeys
2	9 reusable helper flows	Shared onboarding, navigation, and setup logic
3	Jenkins pipeline	Automated execution with HTML reports and video proof
4	BrowserStack integration	Real device testing on Samsung Galaxy S24
5	HTML test reports	Priority badges, module tags, execution logs, video recordings

Recommendations

1. **Expand device coverage** -- Add Pixel 8, OnePlus, and older Android versions on BrowserStack
2. **Schedule nightly runs** -- Automated daily regression on real devices
3. **Release gating** -- Block app releases if P0 tests fail
4. **Fix 3 device-specific assertions** -- Make them flexible for real device differences

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